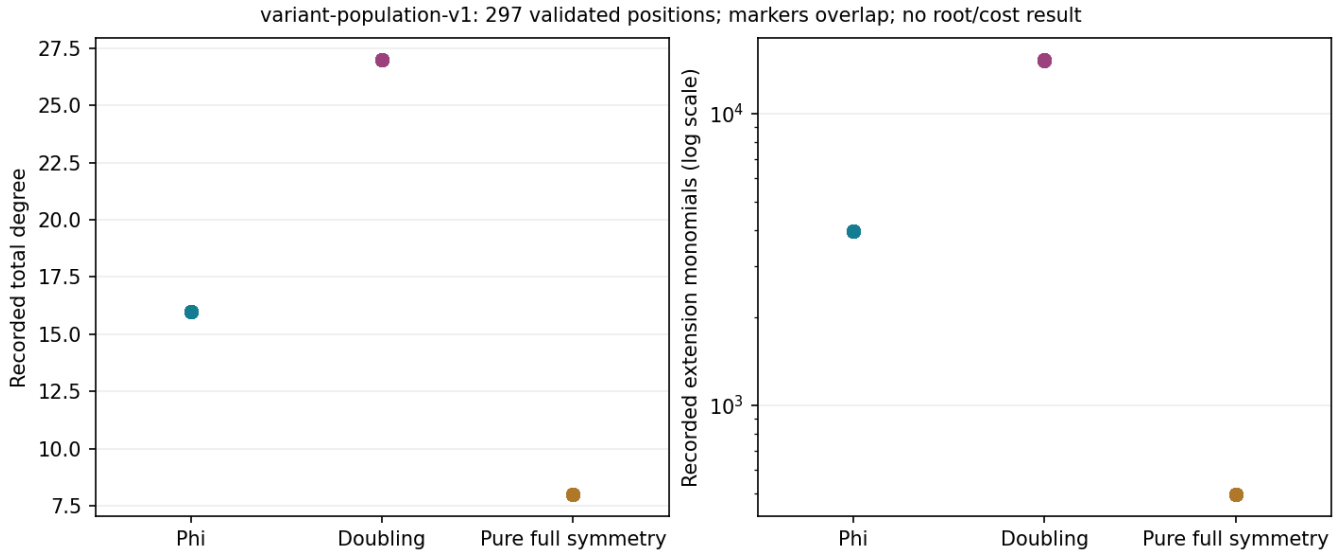
All 297 variant constructions pass native record validation

The managed variant-population-v1 sweep started at 03:20:23 UTC on 2026-10-24. It covers every position of the frozen algebra-v1 workload: seven fields, three seeds, three arms, fixed query positions and labelled positive/regression controls. All 297 requested positions are terminal: 294 new exports and three reused exports from the already completed verification-v8 regression. No solver or relation oracle is repeated.

The independent variant-validation-v1 runner finishes at 06:29:08 UTC and reports pass, with no bounded failures, invalid records or missing positions. Its full native output is variant-population-v1-validation.json (../../results/variant-population-v1-validation.json). The copied receipt is checked byte-for-byte against the terminal native output; its actual SHA-256 is recorded with the publication evidence. Both files hash to 8f20e187ebcb85f753c9fe67a2313471bf1e7fb8303ffbb45afc0c3b86b19da3.



What the validator checks

The exporter is the frozen verification-v8 runner from dc19cc173a457dd4595459765a2fb37a9b02e383. The separate validator is built from fb1733b0ed002e5b8314e41597d6ee03904f0e25; its original gate passes 76 tests before the runner is copied and sealed. It checks the workload, public-input hashes, variable arities, coefficient ranges, equation statistics and hashes, and exact msolve serialization. It checks both finite and geometric exports at every successful position.

The v8 input files omit recorded equation-digest/statistics fields that the original algebra-v1 format carries. The validator computes them, checks the finite statistics against the producer summary, records that those optional input seals are absent, and checks any recorded seal when present. It does not invent missing producer seals. Producer map/symmetry check counts are retained, rather than re-executed in this postprocessor.

Representation sizes and fair controls

The all-position plot shows recorded degrees and extension-monomial counts. Markers can overlap; they are not independent random-curve samples. The p37/seed0/query2 regression has these native recorded values:

Representation	Total degree	Extension monomials
Phi with inverse coordinate, reduced	16	3,960
Doubling with inverse coordinate, reduced	27	15,180
Pure L0 with full four-factor symmetry	8	495

For the same fixture, unreduced phi has 7,425 monomials at degree 16; unreduced doubling has 48,510 at degree 96; partial pure symmetry has 1,485 at degree 16. The recorded reductions change the encoding. They do not measure solving work, regularity, physical relation yield or rank. These numbers describe the implemented encodings, rather than an intrinsic lower bound for every doubling representation.

The pure control is necessary: its permutation symmetry gives a smaller representation too. Phi's inverse-coordinate relation differs from doubling's norm-denominator relation, but an input-size difference alone does not establish a twist-specific algorithmic speedup. The selected native regression already records allocation failure for both mixed implicit encodings under the same 8 GiB cap; those failures remain intact.

Budgets and unresolved measurements

Every new constructor uses one Rust worker, 8 GiB and its original 1,200 s outer watchdog. One busy reservation per field covers the frozen position loop; each export retains its individual limit and receipt. Compilation is separate. Validation uses its frozen native runner, 8 GiB, the shared busy reservation and a 1,200 s watchdog. No elapsed speed comparison is admitted on this shared host.

This is a record/coefficient/export consistency result. It is not an independent generic S5 ideal derivation, a complete-root certificate, a full-stream physical-set/rank result, or an end-to-end price. Those requirements remain open. The raw outputs are outside the earlier 22-directory checkpoint and must enter a later canonical archive version. No leaderboard or benchmark-index result is promoted.